Conducting an Incident Response Investigation (5e)

Digital Forensics, Investigation, and Response, Fifth Edition - Lab 04

Introduction

Digital forensic procedures are integral to incident response investigations. Incident response is a structured approach to identifying, addressing, and mitigating security breaches or cyberattacks, commonly referred to as IT incidents, computer incidents, or security incidents. The primary objective is to handle the situation in a way that limits damage, reduces recovery time, and limits operational costs. While different sources may categorize incident response phases differently, a typical framework consists of six key steps:

1. **Preparation**: An incident response is not a calm and leisurely affair. Without good preparation, an incident response will be disorganized and has the potential to make the incident worse. A formal incident response plan is a crucial first step in preparation. Once an approved plan is in place with the appropriate staffing, ensure that the people assigned incident response duties are properly trained in the processes and procedures necessary for investigating an incident. Additionally, tools such as forensic hardware and software should be acquired and incorporated into the process.

2. **Detection**: This is often a complex endeavor. The detection phase is the part of the incident response process in which the organization first becomes aware of a set of events that possibly indicate malicious activity or constitute a threat to confidentiality, integrity, or availability. Detection can come from internal or external sources, or both.

3. **Analysis**: In this step, personnel begin collecting evidence from systems, such as running memory, log files, network connections, and running software processes. Depending on the type of incident, this collection can take as little as a few hours to several days. The goal of the analysis step is to determine the root cause of the incident and to reconstruct the actions of the threat actor from initial compromise to detection. In this lab, you will use NetWitness Investigator to analyze network traffic and Paraben’s E3 to examine disk images. These tools will help you practice real-world forensic techniques, reinforcing your understanding of incident response analysis.

4. **Containment**: With an understanding of what the incident is and what systems are involved, the next step is the containment phase. This is when the incident response team takes measures to limit the ability of a threat actor to continue compromising network resources, communicating with command-and-control infrastructures, or exfiltrating confidential data. Containment strategies range from locking down ports and IP addresses on a firewall to simply removing the network cable from the back of an infected machine. Depending on who you ask, containment is sometimes considered the first step in the incident response process. While this may be true in terms of overall priorities (an incident should be contained as quickly as possible), in practice, detection, analysis, and containment often overlap and fold back into one another.

5. **Eradication and Recovery**: This step is when the threat actor is removed from the impacted network. In the case of a malware infection, the incident response team may run an enhanced anti-malware solution. Other times, infected machines may be wiped and reimaged. Other activities include removing or changing compromised user accounts. If the team has identified a vulnerability that was exploited, vendor patches are applied or software updates are made. Recovery activities are very closely aligned with those that may be found in an organization's business continuity or disaster recovery plans.

6. **Post-incident Activity**: Post-incident activity includes a complete review of all the actions taken during the incident. The team should discuss what worked and, more importantly, what did not work. It is during

this phase of the process that a written report is completed. For forensic investigators, this type of documentation is critical should the incident ever wind up in court. The documentation should be detailed and show a clear chain of events with a focus on the root cause, if it was determined.

In this lab, you will simulate the Analysis step of the incident response process. You will analyze a packet capture (PCAP) file and disk image for evidence and then prepare your findings in an incident response report. You will then continue your investigation and update your report as more evidence comes to light, mirroring the iterative nature of real-world investigations. By leveraging industry-standard forensic tools, you will bridge the gap between theoretical concepts and hands-on digital forensics practices, preparing you for real-world cybersecurity challenges.

Lab Overview

SECTION 1 of this lab has three parts, which should be completed in the order specified.

1. In the first part of the lab, you will analyze a PCAP file taken during a recent incident with NetWitness Investigator.

2. In the second part of the lab, you will use E3 to identify forensic evidence related to the incident.

3. In the third part of the lab, you will prepare an incident response report.

SECTION 2 of this lab allows you to apply what you learned in **SECTION 1** with less guidance and different deliverables, as well as some expanded tasks and alternative methods.

1. You will analyze additional evidence and update the incident response report accordingly.

Finally, you will explore the virtual environment on your own in **SECTION 3** of this lab to answer a set of questions and challenges that allow you to use the skills you learned in the lab to conduct independent, unguided work—similar to what you will encounter in a real-world situation.

Learning Objectives

Upon completing this lab, you will be able to:

1. Perform forensic analysis as part of an incident response investigation.

2. Perform forensic analysis on a PCAP file using NetWitness Investigator.

3. Perform forensic analysis on a disk image using Paraben’s E3.
4. Correlate evidence from multiple sources to develop your case.
5. Create an incident response report to document your findings.

Topology

This lab contains the following virtual machines. Please refer to the network topology diagram below.

- vWorkstation (Windows: Server 2019)



Tools and Software

The following software and/or utilities are required to complete this lab. Students are encouraged to explore the Internet to learn more about the products and tools used in this lab.

- NetWitness Investigator
- Paraben's E3

Deliverables

Upon completion of this lab, you are required to provide the following deliverables to your instructor:

SECTION 1

1. Lab Report file including screen captures of the following:

- Time Graph
- Details of the 2021-Jul-13 15:33:00 session
- Email containing File Transfer Protocol (FTP) credentials and the associated timestamps

2. Any additional information as directed by the lab:

- Completed Incident Report

SECTION 2

1. Lab Report file including screen captures of the following:

- Email from Dr. Evil demanding that Marvin install a keylogger
- Email from Dr. Evil reminding Marvin to update the firewall and scheduler
- Registry key value associated with the keylogger and the localSPM service

2. Any additional information as directed by the lab:

- Document the Author and Date values associated with the scheduled keylogger task.
- Document the port used for inbound connections to the keylogger and the name and location of the keylogger executable.
- Record the first time and last time the keylogger was started.
- Record whether Marvin interacted with or simply opened the keylogger.
- Updated Incident Report

SECTION 3

1. Lab Report file including screen captures of the following:

- Exfiltrated files in Marvin's Outlook database
- Email with instructions for installing additional spyware

2. Any additional information as directed by the lab:

- None

Hands-On Demonstration

Part 1: Analyze a PCAP File for Forensic Evidence

The setting for this lab is a recent security incident at the game development studio Giggly Goofo. The security incident was discovered on July 31, 2021, at 10:30 AM Eastern time and was reported 10 minutes later. Based on preliminary findings, the security team suspects that a data breach involving both internal and external actors occurred, wherein copies of confidential files were exfiltrated from the corporate network. This distinction is crucial, as internal threats may involve employees misusing access privileges, while external threats often exploit vulnerabilities to infiltrate the network. Understanding the involvement of both helps investigators determine **how** the breach occurred, whether **insider collaboration** played a role, and what mitigation strategies are necessary. As a digital forensics specialist assigned to the incident response team, you have been tasked with performing the Analysis step of the incident response process. Two primary sources of forensic evidence have been provided:

- A PCAP file capturing network traffic at the time of the breach
- A disk image from the system of an employee suspected to be involved in the incident

Since data breaches often involve unauthorized data transfers, analyzing network traffic is a critical first step in understanding the scope of the attack. Network logs provide insight into suspicious connections, data exfiltration patterns, and potential attacker movements within the network.

In this part of the lab, you will analyze the contents of the PCAP file using NetWitness Investigator. Developed by RSA, NetWitness Investigator is an enterprise-level threat analysis platform designed to contextualize network activity and efficiently identify malicious activity. In this lab, you will be using the free version of NetWitness Investigator. While it lacks some advanced enterprise functionalities, it remains a valuable forensic tool for network packet analysis. It enables you to do the following:

- Identify suspicious IP addresses communicating with the corporate network.
- Analyze data exfiltration attempts, such as unauthorized file transfers.
- Track attacker movements based on packet metadata and protocol usage.

In the next steps, you will open the NetWitness Investigator application and import the PCAP file provided by your colleagues on the incident response team.

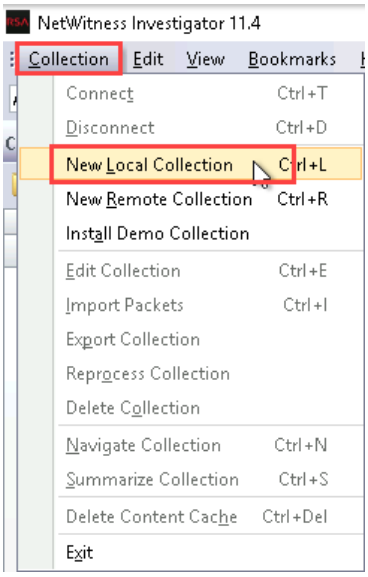
1. On the vWorkstation desktop, **double-click** the **NetWitness Investigator icon** to open the NetWitness application. If you receive a **File Download - Security Warning** dialog box for an http_500_webOC file, please select **Cancel**.



NetWitness Investigator icon

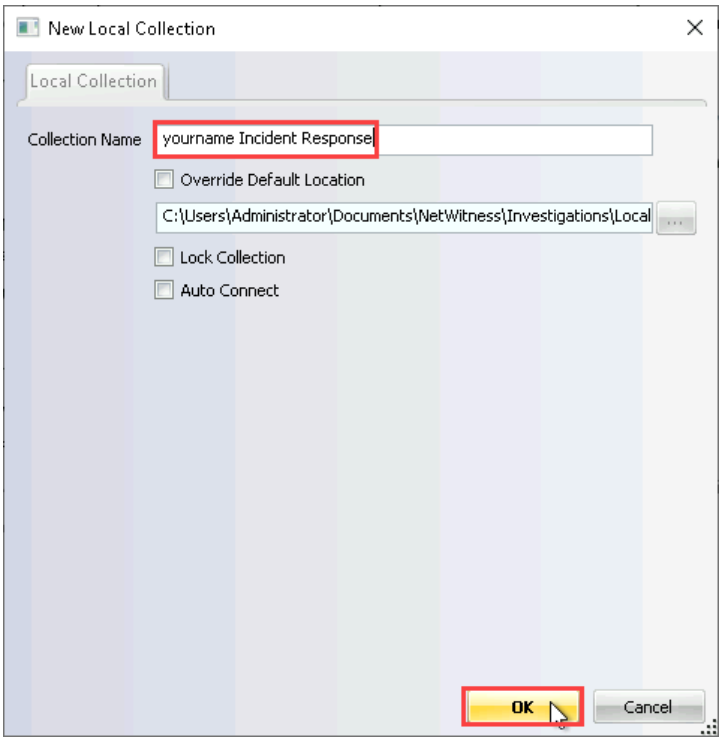
Note: NetWitness Investigator 11.4 uses a legacy Internet Explorer–based component to load its startup Welcome content. On modern versions of Windows, Internet Explorer is deprecated, causing that embedded component to fail and trigger a generic Windows file-download warning. The pop-up is a Windows side effect of this legacy behavior and does not indicate a problem with the lab or the tool.

2. From the NetWitness Investigator menu bar, **click Collection** and **select New Local Collection** from the menu to open the New Local Collection dialog box.



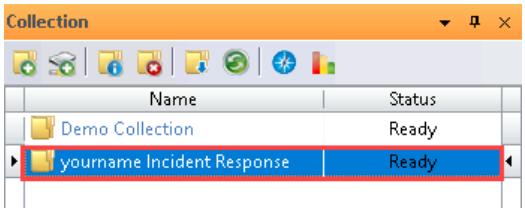
Collection > New Local Collection

3. In the New Local Collection dialog box, **type** ***yourname*** **`Incident Response`** in the Collection Name box, replacing *yourname* with your own name, and **click OK** to name the new collection and close the dialog box.



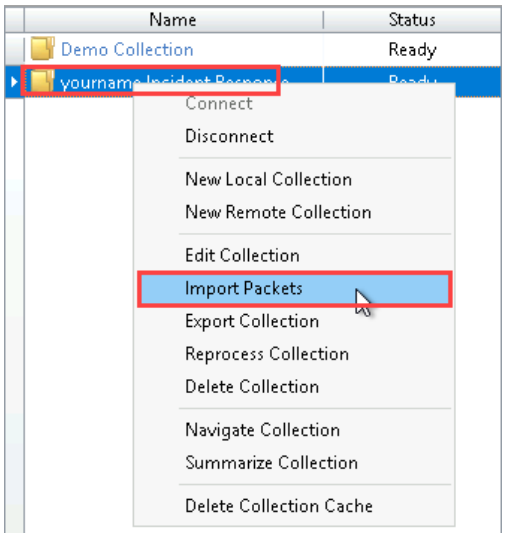
New Local Collection dialog box

4. In the Collection pane, **double-click** the ***yourname*** **Incident Response collection** to activate it and change the status to Ready.



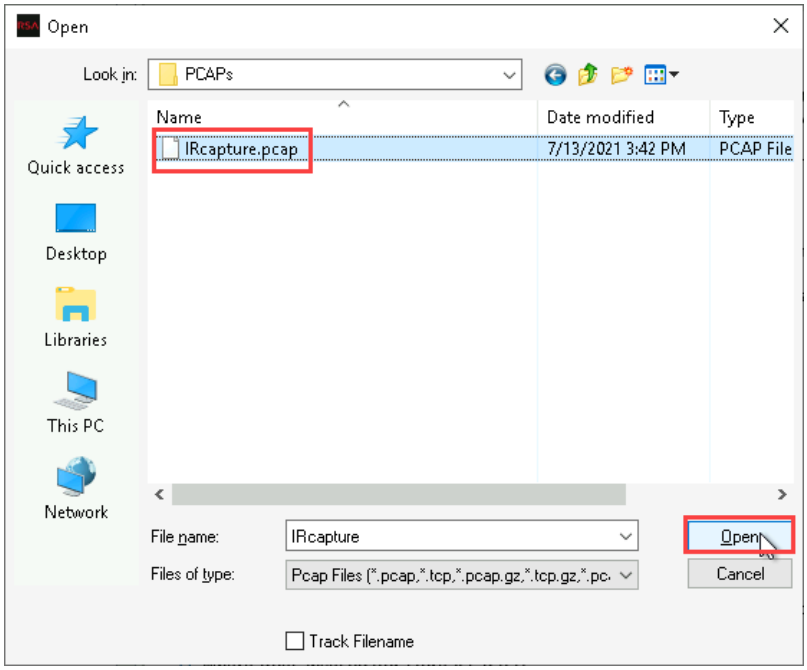
Activate the collection

5. In the Collection pane, **right-click** the ***yourname*** **Incident Response collection** and **select Import Packets** from the context menu to open the Open dialog box.



Import Packets

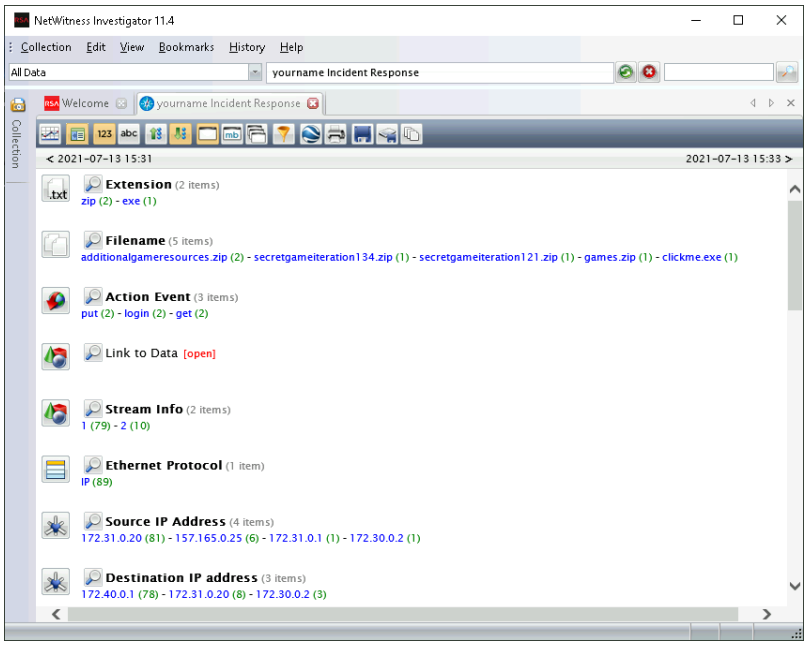
6. In the Open dialog box, **navigate** to **This PC > Local Disk (C:) > Incident Response Evidence > PCAPs** and **select** the **IRcapture.pcap file**, then **click Open** to import the contents of the file into the *yourname* Incident Response Collection.



Open dialog box

Note: While importing, the Status column changes to Importing. When the collection is ready for analysis, the Status column will change to Ready.

7. When the file has finished importing, **double-click** the ***yourname* Incident Response collection** to open it in the Navigation View.

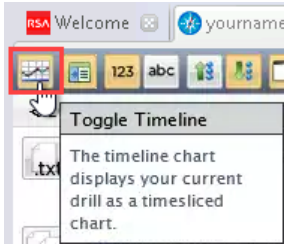


Navigation View

Note: In NetWitness Investigator, collections are explored using the Navigation View. The Navigation View contains multiple reports that are displayed within the Navigation View as individual headers, such as Alerts, Service Type, Source IP Addresses, and more.

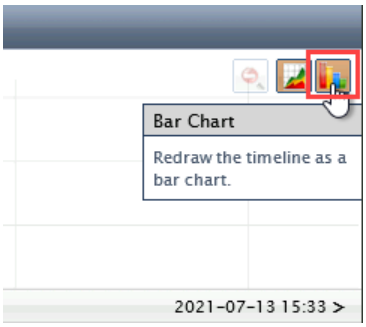
In the top pane, below the Navigation Toolbar, you should see two timestamps. These correspond to the time and date that the packet capture session in the PCAP file began (July 13, 2021, at 3:31 PM) and ended (July 13, 2021, at 3:33 PM).

8. In the Navigation View, **click** the **Toggle Timeline button** to display the Time Graph.



Toggle Timeline

9. In the Time Graph, **click** the **Bar Chart button** to display the time graph as a bar chart.



Bar Chart

Note: The Time Graph provides a breakdown of the sessions contained in the PCAP file. In the Bar Chart display mode, you can hover your cursor over a specific bar to see how many sessions occurred in that interval. In both the Area Chart and Bar Chart display modes, you can drag and drop your cursor within any blue area to zoom in and narrow the focus of NetWitness to sessions that occurred within the selected interval. For example, if you had reason to suspect that malicious network activity occurred at 10:07 AM (perhaps an employee clicked a malicious link in a phishing email), you could use the zoom feature to display only the NetWitness reports related to traffic occurring at that exact moment. To zoom back out at any time, you can click the Zoom Out Timeline button next to the Area Chart and Bar Chart buttons.

10. **Make a screen capture** showing the **Time Graph**.

11. **Click** the **Toggle Timeline button** to hide the Time Graph.

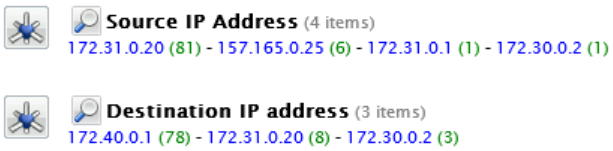
12. In the Navigation View, **locate** the **Filename report**.



Filename report

Note: The Filename report lists any files that were detected during the capture session. Considering that your employer is a game developer and the security incident involved the exfiltration of confidential company data, the four .zip files listed here might contain the stolen data. The fifth file, clickme.exe, does not appear to be related to game development, but it could be a malicious executable.

13. In the Navigation View, **locate** the **Source IP address** and **Destination IP address reports**.



IP Address reports

Note: The Source IP Address report contains IP addresses from which connections were established in the PCAP file. Similarly, the Destination IP Address report contains IP Addresses to which connections were established. You should see four entries under Source IP Address and three entries under Destination IP Address. Altogether, you should see four unique IP addresses, three of which share portions of the same network address (172.x.x.x). As an IT employee at Giggly Goofo, you know that all internal IP addresses on the internal company network begin with 172. Therefore, your attention should immediately be drawn to 157.165.0.25. The PCAP was taken within the Giggly Goofo LAN, and this IP address appears to be an external device connecting to the LAN, which makes it a strong candidate for further investigation.

14. Under the Source IP Address report, **click** the **(6)** next to **157.165.0.25** to open the Session List View for this source IP address.

Note: As you drill down into specific reports, NetWitness keeps track of your current location in the Drill Path at the top. If you need to return to the Navigation View for the yourname Incident Response collection at any time, click the yourname Incident Response link within the Drill Path.

15. In the Session List View for 157.165.0.25, **locate** the **session that started on 2021-Jul-13 15:33:00**. This is the earliest session of the series, with the largest number of events.

Note: In the detailed session information on the right, you should see the names of two of the files that you previously identified as potential exfiltrated data. According to the session details, an FTP transfer occurred between 157.165.0.25 and 172.31.0.20 using the username GigglyGoofoDev and password Il0veC0d!nG. This appears to be at least one source of the data breach and provides valuable information for continuing your investigation, including IP addresses, credentials, timestamps, and filenames. Naturally, you will need to document all these details for later use in your incident response report.

16. **Make a screen capture** showing the **details of the 2021-Jul-13 15:33:00 session**.

17. **Close** the **NetWitness Investigator window**.

Part 2: Analyze a Disk Image for Forensic Evidence

In Section 1, Part 1, you conducted a network traffic analysis to identify potential indicators of compromise. While network logs provided insights into suspicious activity, they only tell part of the story. To build a stronger case, forensic investigators often turn to disk analysis, which can reveal deleted files, traces of malware, unauthorized data access, and other critical evidence that may not be visible in network logs alone.

In this part of the lab, you will analyze a disk image obtained from the laptop of Marvin Jonson, an employee at Giggly Goofo. The security team flagged Marvin as a person of interest after network logs suggested potential involvement in a data exfiltration incident. Examining his disk image may uncover crucial evidence, such as files that were exfiltrated, unauthorized software installations, communication logs, or other forensic artifacts linking him to the security breach.

To perform this analysis, you will use Paraben's E3, a digital forensic tool designed for comprehensive disk analysis. Paraben's E3 allows investigators to recover deleted files, analyze metadata, examine file system activity, and detect suspicious artifacts that may indicate data theft or malicious intent.

In the next steps, you will:

- Import the disk image into Paraben's E3 to access its contents.
- Examine the file system and search for forensic artifacts, including deleted files, recently accessed documents, unauthorized software, and potential traces of exfiltrated data.
- Correlate findings with network activity from Part 1 to build a clearer picture of how the security incident unfolded.

By analyzing both network and disk-based evidence, you will practice a core principle of incident response: correlating multiple sources of forensic data to reconstruct an event and identify those responsible.

1. On the vWorkstation desktop, **double-click** the **Electronic Evidence Examiner icon** to open the E3 application



E3 icon

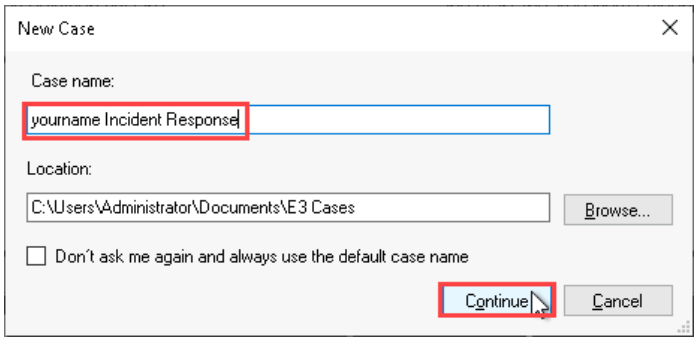
Note: E3 may take several minutes to load. The E3 welcome screen opens with shortcuts to the most common activities that forensic investigators will perform within the tool.

2. On the Welcome screen, **click** the **Add Evidence button** to open the New Case dialog box.



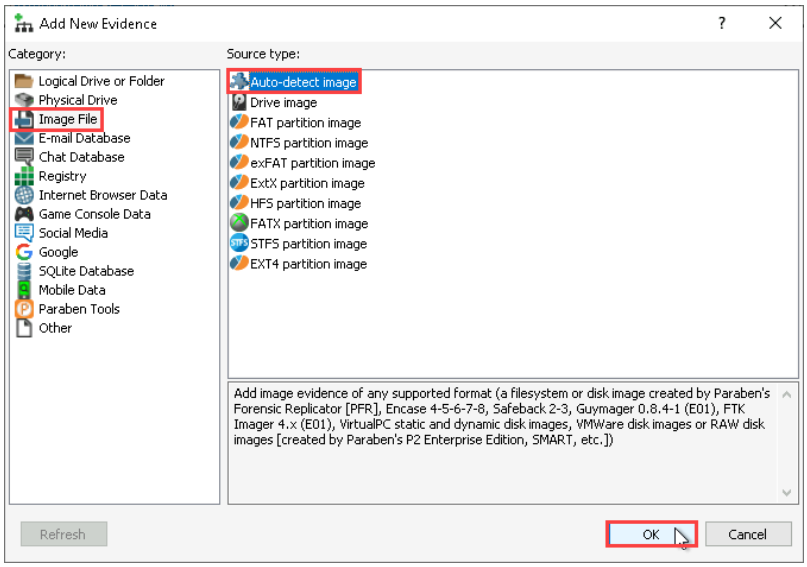
Welcome page - Add Evidence

3. In the New Case dialog box, **type** ***yourname*** **Incident Response** in the Case name field, replacing *yourname* with your own name, then **click Continue** to create your new case file and open the Add New Evidence dialog box.



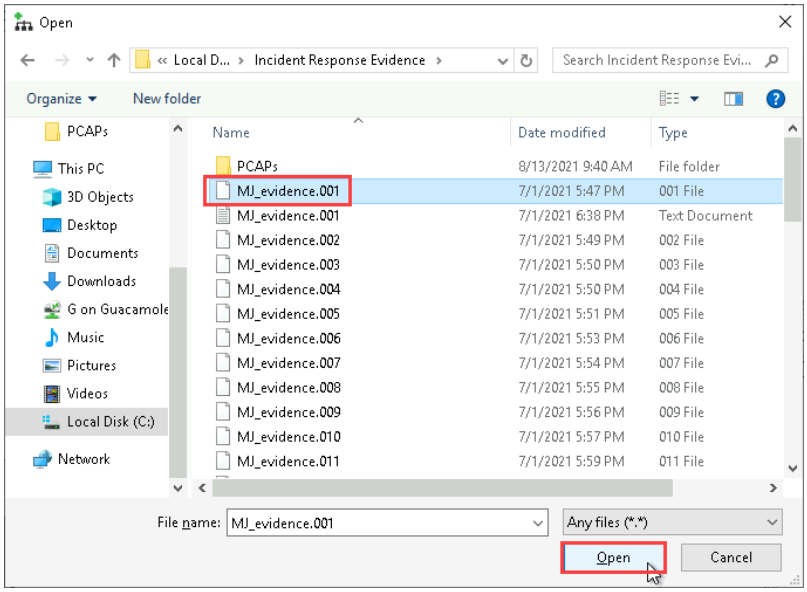
New Case dialog box

4. In the Add New Evidence dialog box, **click** the **Image File category**, then **select** the **Auto-detect image Source type** and **click OK** to continue.



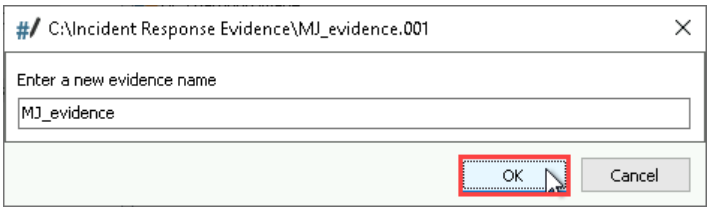
Add New Evidence - Auto-detect image

5. In the Open dialog box, **navigate** to **This PC > Local Disk (C:) > Incident Response Evidence** and **select** the first **MJ_evidence.001 file**, then **click Open** to import the digital drive image for this lab.



Open dialog box

6. When prompted, **click OK** to accept the default name for the drive image and add the data from the drive image to your case file.

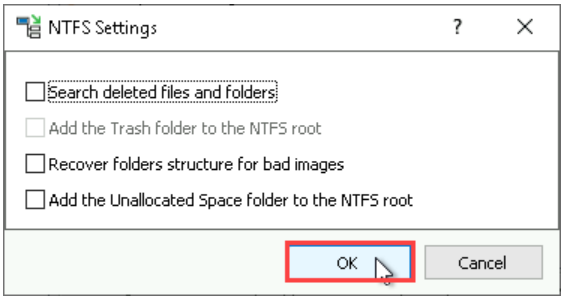


Evidence name

Note: The yourname Incident Response case will appear in the Case Content pane. The Case Content pane is the primary display and navigation area for all evidence in the open case file. Within the Case Content pane's tree-view structure, you can access case nodes, evidence nodes, evidence type nodes, folder nodes, and data grids/streams. Technically, the Case Content pane does not store the actual evidence but provides a series of links to elements within the physical evidence.

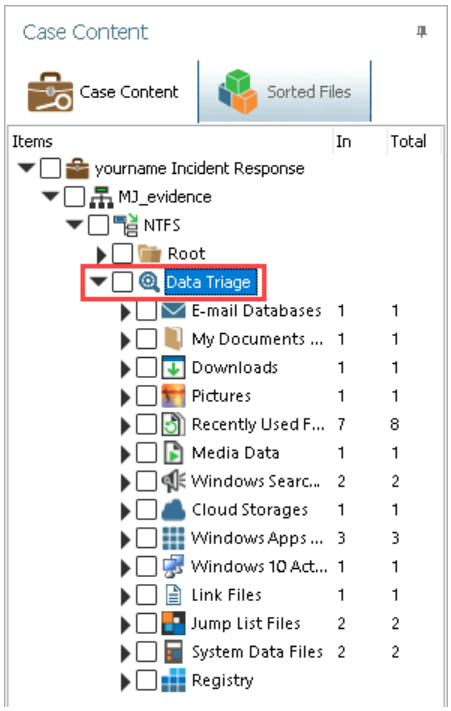
When you select a specific node or grid within the Case Content pane, the contents of the enclosing folder or database will be displayed in the Data Viewer in the center pane. Within the Data Viewer, you can select individual files, folders, and records. Additional information about the currently selected node, grid, file, folder, or record will be displayed in the Viewers pane on the right, which provides multiple ways to view your current selection.

7. When prompted, **click OK** to close the NTFS Settings dialog box without making any changes.



NTFS Settings dialog box

8. In the Case Content pane, **navigate** to **yourname Incident Response / MJ_evidence / NTFS** and then **expand** the **Data Triage node** to display the Data Triage categories.

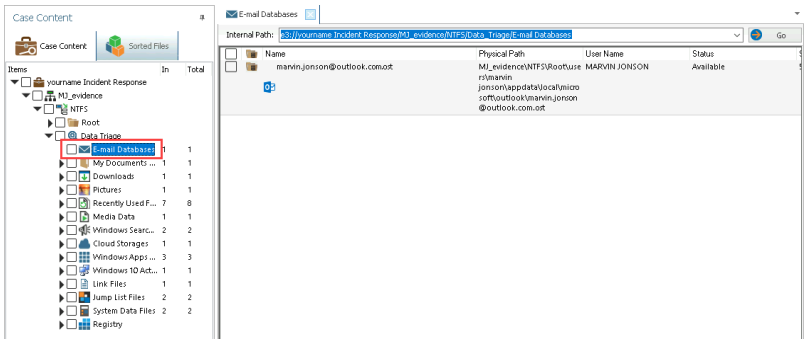


Data Triage

Note: The Data Triage function in E3 allows you to quickly view potentially high-value evidence, such as email databases, recently used files, and parsed registry data.

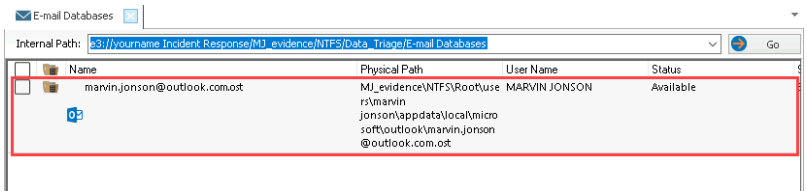
In the next steps, you will use E3's Advanced Search functionality to search for evidence in Marvin's email records.

9. In the Case Content pane, **select** the **E-mail Databases category** to display all email databases detected on the drive image in the Data Viewer.



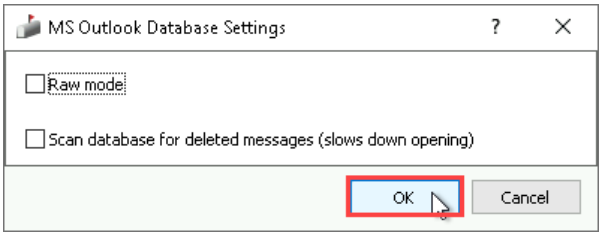
E-mail Databases category

10. In the Data Viewer, **double-click** the **marvin.jonson@outlook.com.ost database** to open it.



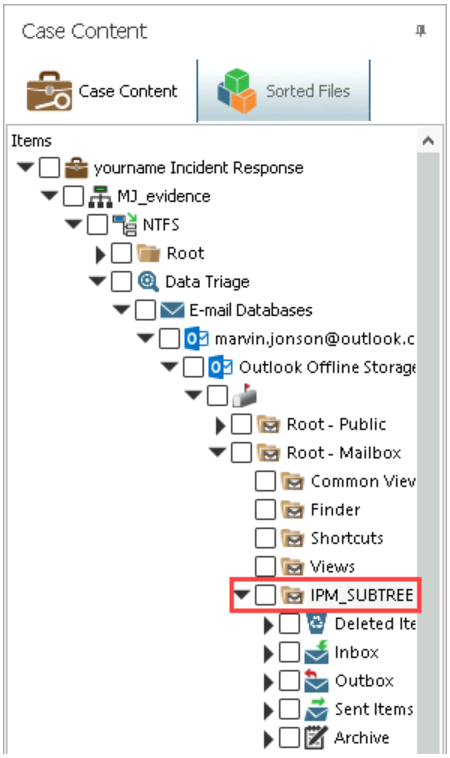
Marvin.jonson@outlook.com.ost database

11. When prompted, **click OK** to close the MS Outlook Database Settings dialog box without making any changes.



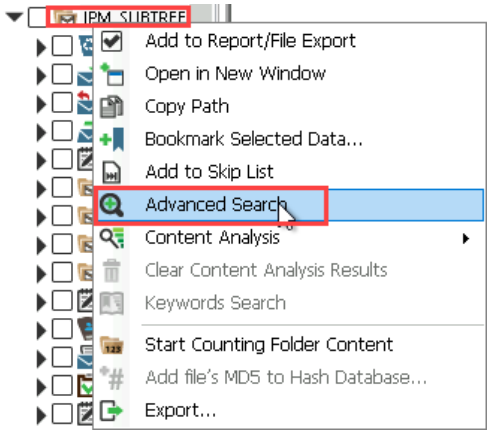
MS Outlook Database Settings dialog box

12. In the Case Content pane, **navigate to marvin.jonson@outlook.com.ost\Outlook Offline Storage\Root - Mailbox** and then **expand** the **IPM_SUBTREE folder node** to display the structure of the email database.



IPM_SUBTREE

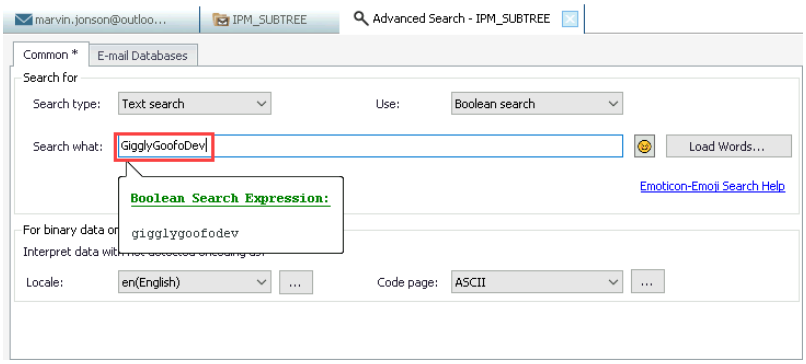
13. In the Case Content pane, **right-click** the **IPM_SUBTREE folder node** and **select Advanced Search** from the context menu to open the Advanced Search pane in the center console.



Advanced Search

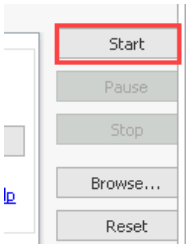
Note: From your work in Section 1, Part 1, you know that data was exfiltrated directly from the Giggly Goofo FTP server using compromised credentials. If Marvin was knowingly involved in the security breach, it seems likely that he may have shared the FTP credentials via email. For that reason, running a search for the FTP username seems like a good start.

14. In the Advanced Search pane, **type `GigglyGoofoDev`** in the Search what field.



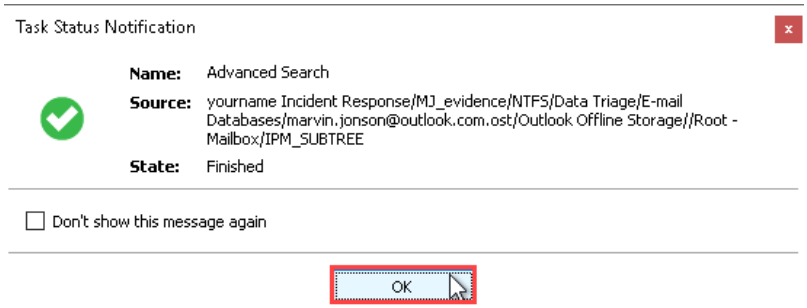
Search what field

15. **Click** the **Start button** to run the search. You may need to scroll to the right to see the Start button. The search will take about 1 minute to complete.



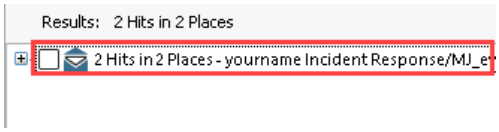
Start button

16. When prompted, **click OK** to close the Task Status Notification dialog box.



Task Status Notification dialog box

17. In the Results pane, **double-click** the **search results** to display the email in the E-mail Data pane and open the enclosing folder in a new Data Viewer tab. You may need to extend the E-mail Data pane to see the full email.



Results

Note: And there it is! Your search should return an email in which Marvin appears to send the FTP server's publicly facing IP address and valid credentials to one Dr. Evil. While the nature of the relationship between these two men arguably raises more questions than answers, at this point you have actionable proof that Marvin is involved in the data theft incident.

18. **Make a screen capture** showing the **email containing FTP credentials and the associated timestamps**.

19. **Close** the **E3 window**.

Part 3: Prepare an Incident Response Report

In this part of the lab, you will transition from forensic analysis to documenting your findings in an incident response report. Clear and structured documentation is a critical component of digital forensics and incident response, as it helps organizations reconstruct security incidents, strengthen defenses, and support legal proceedings when necessary. A well-documented report ensures that key stakeholders, including security teams, management, and, in some cases, law enforcement, can understand the scope and impact of the incident.

To facilitate this process, you will reference a standardized Incident Reporting Template developed by Carnegie Mellon University for the United States Computer Emergency Readiness Team (US-CERT). Standardized templates ensure consistency in reporting, facilitate communication between response teams, and help correlate evidence from multiple sources, such as network captures (PCAP files) and disk images. This synthesis of findings is essential for identifying patterns, proving intent, and constructing an accurate timeline of the breach.

1. **Navigate to** the **Incident Management Guide** at https://us-cert.cisa.gov/sites/default/files/c3vp/crr_resources_guides/CRR_Resource_Guide-IM.pdf and **review** the **"Example Incident Reporting Template"** on page 42.

Note: Using this template, you will compile an abbreviated incident report based on the evidence you collected in Parts 1 and 2. This structured approach will help reinforce your forensic reporting skills and highlight how correlating evidence strengthens an investigation.

2. **Review** the information you gathered in Parts 1 and 2 of this section.

3. **Complete** the following incident response template.

Giggly Goofo Game Studios

Incident Response Team - Incident Reporting Template

Date
Insert current date here.

Name
Insert your name here.

Incident Priority

Define this incident as High, Medium, Low, or Other.

Incident Type

Include all that apply: Compromised System, Compromised User Credentials, Network Attack (e.g., Denial-of-Service), Malware (e.g., virus, worm, trojan), Reconnaissance (e.g., scanning, sniffing), Lost Equipment/Theft, Physical Break-In, Social Engineering, Law Enforcement Request, Policy Violation, Unknown/Other.

Incident Timeline

Define the following: Date and time when the incident was discovered, date and time when the incident was reported, and date and time when the incident occurred, as well as any other relevant timeline details.

Incident Scope

Define the following: Estimated quantity of systems affected, estimated quantity of users affected, third parties involved or affected, as well as any other relevant scoping information.

Systems Affected by the Incident

Define the following: Attack sources (e.g., IP address, port), attack destinations (e.g., IP address, port), IP addresses of the affected systems, primary functions of the affected systems (e.g., web server, domain controller).

Users Affected by the Incident

Define the following: Names and job titles of the affected users.

Note: This concludes Section 1 of the lab.

Applied Learning

Note: SECTION 2 of this lab allows you to apply what you learned in SECTION 1 with less guidance and different deliverables, as well as some expanded tasks and alternative methods. You will analyze additional evidence and update the incident response report accordingly.

Part 1: Identify Additional Email Evidence

After successfully completing an initial incident response report and submitting it to the rest of the incident response team and senior leadership for review, the security team quickly isolated the FTP server and locked all of Marvin Jonson's accounts. While your peers have commended you for your quick response and detailed documentation, you are concerned that your initial findings do not encompass the full scope of the incident and that additional data may have been exfiltrated through other means. After all, attackers often employ multiple exfiltration methods, including email, cloud storage, and removable media, to ensure the success of their

operation. If Marvin was so easily compelled to give up access to the company FTP server, what else might he have used to remove sensitive data?

To further your investigation, you will analyze Marvin's email archives for additional evidence. Email is a critical forensic artifact, often preserving logs of communication, timestamps, and links to external actors. Deleted messages, attachments, or traces of suspicious conversations may provide deeper insight into Marvin's actions. Pay close attention to email headers and metadata, which can be just as revealing as the content itself. By correlating email-based evidence with previous findings, you will refine your understanding of the incident and strengthen your forensic analysis skills.

1. From the vWorkstation desktop, **launch** the **E3 application**.

2. From the Welcome screen, **open** the ***yourname*** **Incident Response case file** that you created in Section 1.

3. In the Case Content pane, **expand** the **Data Triage node** and then **select** the **E-mail Databases category** to display the contents in the center pane.

4. In the center pane, **double-click** the **marvin.jonson@outlook.com.ost database** to open it.

5. In the Case Content pane, **navigate to marvin.jonson@outlook.com.ost\Outlook Offline Storage\Root - Mailbox** and then **expand** the **IPM_SUBTREE folder**.

6. In the Case Content pane, **right-click** the **IPM_SUBTREE folder** and **select Advanced Search** to open an Advanced Search pane in the center console.

7. In the Advanced Search pane, **click** the **Email Databases tab**, then **click** the **Sender(From) checkbox** and **type** `evildr683@yahoo.com`. You may need to scroll to the right to see the Filter sender and recipient section.

8. **Click Start** to run the search for emails from Dr. Evil. You may need to scroll to the right to see the Start button.

9. When the search is complete, **review** the results.

Note: As you review the email exchange between Marvin and Dr. Evil, you wonder exactly how Marvin met this strange man and what leverage Dr. Evil has over him, but those details are well outside the scope of your immediate investigation. Within the results, you should see evidence that Dr. Evil instructed Marvin to install a keylogger and then make changes to both the firewall and scheduled tasks. Both demands should be cause for concern and will require further investigation.

10. **Make a screen capture** showing the **email from Dr. Evil demanding that Marvin install a keylogger.**

11. **Make a screen capture** showing the **email from Dr. Evil reminding Marvin to update the firewall and scheduler.**

Part 2: Identify Evidence of Spyware

You now have reason to believe that Marvin may have installed a keylogger on his company workstation and made unauthorized changes to the firewall and scheduled tasks. A keylogger is a serious security threat because it records every keystroke made on the system, including usernames, passwords, and other sensitive information, potentially enabling credential theft and unauthorized access. Detecting and analyzing keylogger activity is a critical part of digital forensics for understanding the extent of a security breach. When conducting a forensic investigation on a Windows drive image like this one and attempting to identify evidence of application installations or changes to system functions, one of your first stops should be the Windows Registry. The Windows Registry is a database that stores a wide range of settings for the operating system and applications installed on it, as well as user activity. Even after an application is removed, forensic artifacts such as auto-run keys, installation logs, and modification timestamps can reveal traces of its presence. These registry entries can help establish a timeline of when Marvin installed or used the suspected spyware.

Attackers often modify system settings like the firewall and scheduled tasks to maintain persistence and evade detection. Altering firewall rules can allow malicious programs to communicate externally without triggering security alerts, while scheduled tasks can be leveraged to automatically execute malware upon system startup. Understanding how these techniques are used will help correlate evidence across multiple sources.

In the next steps, you will open the Registry category in the Data Triage view and run a search for registry settings related to keylogging software. Specifically, look for registry keys that indicate persistence mechanisms, such as auto-run entries under HKLM\Software\Microsoft\Windows\CurrentVersion\Run, installation paths stored in HKLM\Software or HKCU\Software, and timestamps that could provide insight into when Marvin installed or modified the software.

1. In the Case Content pane, **navigate to Data Triage / Registry / Windows 10 Enterprise** and then **expand** the **Incident Response node**.

Note: Similar to the Data Triage function, the Incident Response node is an E3 function that automatically aggregates some of the most high-value sources of evidence in the Windows Registry, which allows investigators to quickly surface relevant evidence.

2. In the Incident Response node, **right-click** the **Scheduled Tasks category** and **select Advanced Search** to open an Advanced Search pane in the center console.

3. Use the Advanced Search function to **run a search** for the term **`keylogger.`**

4. When the search is complete, **review** the results.

5. **Document** the Author and Date values associated with the scheduled keylogger task.

6. **Repeat steps 2–4** on the FirewallPolicy category.

7. **Document** the port used for inbound connections to the keylogger and the name and location of the keylogger executable.

8. **Repeat steps 2–4** on the Services category, this time running a search for the name of the keylogger executable.

9. **Make a screen capture** showing the registry key value associated with the keylogger and the localSPM service.

Note: Now that you have concrete proof that Marvin installed a keylogger on his workstation, you will need to learn more about how he used it. To accomplish this, you will explore the SQLite database associated with the Windows 10 Activity Timeline. The Activity Timeline is a feature that was added to Windows 10 in 2018 with the stated intention of helping users remember what they did on a particular day. For forensic investigators, the Timeline is a veritable goldmine of information, as it effectively provides a granular chronological record of every activity that a user performed. In the next steps, you will drill down into Marvin's timeline and search for activity related to the keylogger.

10. In the Case Content pane, **navigate** to **Data Triage / Windows 10 Activity Timeline / ActivitiesCache.db / $DATA / SQLite Database / Tables / Activity**.

11. **Repeat steps 2–4** on the Activity 1-194 table, running a search for the name of the keylogger executable.

12. In the results, **identify** the first and last start times for the keylogger.

Note: When you double-click each search result, the database record will be displayed within its enclosing table in a new Data Viewer pane, while the contents of the record will be displayed in the Properties pane. Within the Properties pane, you should see values identified as AppId, StartTime, and ExpirationTime. If you hover over the AppId value, you should see the full filepath for the application associated with the record, including the name of the keylogger executable. The StartTime and ExpirationTime should identify exactly when the application started and ended. However, they are both recorded as Unix timestamps, which are not particularly legible to humans. In the next steps, you will use a Unix timestamp converter to translate these dates into conventional, human-legible timestamps.

13. From the vWorkstation taskbar, **launch Chrome** and **navigate to** the **converter at https://www.epochconverter.com/**.

14. **Copy and paste** the **first start time and last start time** for the keylogger into the hex converter.

15. **Record** the first time and last time the keylogger was started.

Note: Within the Properties pane, you should also see a value titled ActivityType. Activity Type will contain either a 5 or a 6, where 5 means that the user opened the application and 6 means that they interacted with the application. Using this information, you should be able to identify whether Marvin interacted with the keylogger or simply opened it.

16. In the results, **review** the values in the **ActivityType column**.

17. **Record** whether Marvin interacted with or simply opened the keylogger.

Part 3: Update an Incident Response Report

Incident response investigations are iterative by nature, requiring analysts to continuously refine their reports as new evidence emerges. You now have additional evidence demonstrating that Marvin was compelled by Dr. Evil to install a keylogger and make other unauthorized changes to his workstation. This includes email communications confirming the coercion, as well as concrete proof from the Windows Registry verifying that he carried out these actions. These findings significantly expand the scope of your initial report by providing insight into intent, methods, and affected systems.

In real-world forensic investigations, updating reports is critical for several reasons. From a legal and compliance perspective, maintaining accurate and complete documentation ensures that findings hold up under scrutiny in court or regulatory reviews. From a security decision-making standpoint, updated reports allow stakeholders to respond effectively, mitigating threats based on the most current intelligence. A well-maintained forensic report serves as an evolving record that guides remediation efforts and improves future incident response.

Forensic reporting best practices emphasize clarity, conciseness, and accuracy while maintaining an accurate timeline of events. As you update your report, consider how your new findings integrate with your prior evidence, such as how email logs, registry modifications, and firewall activity correlate to form a more complete picture of the attack. Documenting these connections strengthens your analysis and correlates evidence from multiple sources.

To update your report effectively:

- Identify key changes: What new evidence has emerged? How does it alter your understanding of the incident?
- Incorporate meaningful updates: Focus on significant findings, such as new indicators of compromise, newly identified malicious actors, or additional affected systems.
- Reassess the timeline: How do the new findings impact the sequence of events? Does this change your interpretation of Marvin's role or Dr. Evil's tactics?

Using the incident report template, update your documentation accordingly. If a section remains unchanged from your first report, you may indicate this by responding with *Unchanged*. However, ensure that all relevant updates are fully integrated so that decision-makers have the most accurate and actionable understanding of the situation.

1. Complete the following incident response template. You do not need to repeat any details from Section 1. You just need to incorporate the new information that you discovered in Section 2.

Giggly Goofo Game Studios

Incident Response Team - Incident Reporting Template

Date
Insert current date here.

Name
Insert your name here.

Incident Priority
Has the incident priority changed? If so, define the new priority. Otherwise, state that it is unchanged.

Incident Type
Has the incident type changed? If so, define any new incident type categories that apply. Otherwise, state that it is unchanged.

Incident Timeline
Has the incident timeline changed? If so, define any new events or revisions in the timeline. Otherwise, state that it is unchanged.

Incident Scope
Has the incident scope changed? If so, define any new scoping information. Otherwise, state that it is unchanged.

Systems Affected by the Incident
Has the list of systems affected changed? If so, define any new systems or new information. Otherwise, state that it is unchanged.

Users Affected by the Incident
Has the list of users affected changed? If so, define any new users or new information. Otherwise, state that it is unchanged.

Note: This concludes Section 2 of the lab.

Challenge and Analysis

Note: The following exercises allow independent, unguided work using the skills you learned earlier in this lab, similar to what you would encounter in a real-world investigation.

Part 1: Identify Additional Evidence of Data Exfiltration

Threat actors often employ multiple methods to exfiltrate data, knowing that a single avenue of theft may be detected and blocked. While you have already conducted two thorough investigations, experience has shown that intellectual property theft can involve multiple channels, including cloud storage, USB devices, and email. One common tactic is to send stolen files as email attachments, sometimes obfuscating them within archives or renaming them to evade detection. To ensure that you have accounted for all potential exfiltration methods, you will now shift your focus to Marvin’s email activity.

Specifically, you will examine the attachments stored within Marvin's Outlook account, paying close attention to his Sent folder. Your goal is to identify any files that may contain Giggly Goofo intellectual property. Look for common indicators of exfiltrated data, such as proprietary file types, unusual naming conventions, or metadata inconsistencies that suggest tampering. Additionally, ensure that you document your findings thoroughly, as proper forensic documentation is critical in professional investigations.

By correlating these findings with previously analyzed evidence, you will reinforce the importance of cross-referencing multiple data sources—an essential skill in digital forensics.

Make a screen capture showing an **exfiltrated file in Marvin's Outlook database**.

Part 2: Identify Additional Evidence of Spyware

Upon reviewing previous findings, you notice a suspicious email in Marvin's inbox that appears to be from the Security team. The email references the installation of monitoring software and urges immediate action. Phishing emails are a common method for delivering spyware, often masquerading as legitimate security communications to trick users into installing malicious software.

Recognizing that legitimate security teams do not request software installation via email, you examine the message for indicators of phishing, such as inconsistencies in the sender's address, unexpected urgency, or suspicious links. Given the context of your investigation, this email could be a crucial piece of evidence connecting Marvin’s actions to a broader compromise.

To build a comprehensive forensic case, you should analyze the email headers, sender details, and any attachments or links. Correlate this finding with prior evidence from Part 1, including system logs, disk images, and any signs of unauthorized software installations. Understanding how attackers use phishing to gain initial access can provide insight into how the network may have been further compromised.

Make a screen capture showing the email with instructions for installing additional spyware.

Note: This concludes Section 3 of the lab.